AXIAL ENGINEERS

DELIVERING EXCELLENCE

SCREW CONVEYORS





AXIAL ENGINEERS
DELIVERING EXCELLENCE

OUR PHILOSOPHY

A

A – Accuracy in Engineering

We deliver precision-crafted screw conveyors and material handling solutions tailored to your specific needs — because in engineering, accuracy isn't optional, it's essential.

X

X – Xcellence in Every Detail

From raw material to final product, excellence defines our process. Every weld, every screw, every component reflects our unwavering commitment to quality.

I

I – Innovation with Integrity

We don't just build; we innovate with purpose. Our solutions are born from experience, powered by creativity, and delivered with honesty and trust.

A

A – Agility in Action

Markets shift. Needs change. Axial Engineers moves with speed and flexibility, ensuring we remain a responsive and reliable partner in a dynamic world.

L

L – Long-Term Partnerships

We're not here for transactions — we're here for relationships. With every product and every promise, we build lasting trust that drives mutual growth.



About Axial Engineers

Established in Vatva, Ahmedabad, Axial Engineers has grown into a trusted name in material handling solutions, with a specialization in screw conveyors and over 20 years of manufacturing excellence. Located at B-3, Khodiyar Estate, Vatva GIDC Phase-IV, we are committed to delivering precision-engineered products backed by personalized service.

At Axial Engineers, we offer more than just equipment — we provide dependable solutions with a sharp focus on quality, reliability, and customer satisfaction. Our foundation is built on the belief that consistent service and robust engineering lead to long-term partnerships and industry leadership.

Success in Business – Axial's Philosophy

Success at Axial Engineers stems from strategic thinking, a deep understanding of customer needs, and a sense of ownership at every level. We believe in:

- Taking initiative and personal responsibility
- Fostering a positive, collaborative work culture
- Clear, effective communication
- Continuous innovation and rejection of complacency

By aligning our actions with a greater purpose, we ensure that every team member contributes to our collective success. We stay rooted in the fundamentals: quality, service, and cost-effectiveness, streamlining efforts for maximum efficiency.

THE HEART OF EVERY EFFICIENT

SLUDGE DEWATERING SCREW PRESS

Precision Engineered for Superior Dewatering Performance

The dewatering screw is the core component of every screw press machine. Designed for maximum sludge compression and efficient water separation, Axial Engineers' precision-fabricated screws deliver superior dewatering performance, longer service life, and reliable operation in demanding wastewater treatment applications.

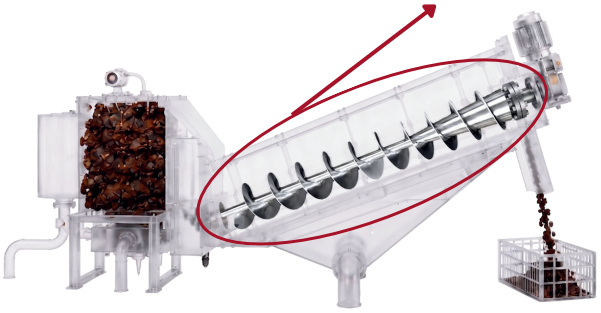
WEAR-RESISTANT FLIGHTS
Built for continuous operation in abrasive conditions

TAPERED COMPRESSION ZONE
Increases sludge compression for improved dewatering

PRECISION MANUFACTURING
Accurate flight geometry for consistent performance

SS304/SS316 CONSTRUCTION
Excellent corrosion resistance and durability





APPLICATIONS

- Municipal Wastewater Treatment
- Industrial Effluent Treatment
- Food Processing Plants
- Chemical Industries
- Paper & Pulp Industries
- Textile Industries

KEY FEATURES

- Precision Fabricated Screw Flights
- SS304 & SS316 Construction
- Optimized Tapered Compression Zone
- Superior Wear & Corrosion Resistance
- Efficient Sludge Conveyance
- Lower Moisture Content in Sludge Cake
- Long Service Life
- Custom Sizes Available

AXIAL ENGINEERS
DELIVERING EXCELLENCE

UNRIVALED OFFERING

OUR PRODUCTS

Engineered for performance. Built for reliability.
Solutions that keep your operations moving.



SCREW CONVEYORS



HORIZONTAL SCREW CONVEYORS

- Most widely used type of screw conveyor
- Available in various sizes, lengths, configurations, and materials
- Ideal for dry to semi-fluid bulk materials
- Multiple inlet and discharge points
- Totally enclosed design

BENEFITS

- Cost-effective compared to other conveying devices
- Efficient bulk material distribution
- Suitable for dusty, corrosive environments

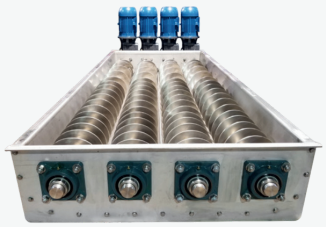


SCREW FEEDERS

- Designed to meter bulk materials
- Capacity and feed rate control with variable speed drives
- Available in various sizes, lengths, configurations, and materials
- Inlet always flood loaded (100%)

BENEFITS

- Accurate metering of bulk materials
- Customizable feed rates
- Suitable for use with hoppers, bins, and silos



LIVE-BOTTOM SCREW FEEDERS

- Designed for large silos, bins, and hoppers
- Multiple feeder screws
- Create a "live bottom"
- Measures and draws materials equally from the entire inlet opening

BENEFITS

- Prevents bridging or cavern formation
- Ideal for bulk materials that compact or bridge easily
- Even distribution of materials



INCLINED SCREW CONVEYOR

- Higher horsepower requirement than horizontal conveyors
- Efficiency varies with incline angle and material characteristics
- Best with tubular/shrouded cover design and few intermediate hanger bearings
- Operates efficiently at high speeds

BENEFITS

- Effective for inclined material transport
- Handles various materials with appropriate design
- High-speed operation minimizes material fallback
- Tailored design for specific needs

MATERIAL HANDLING SOLUTIONS



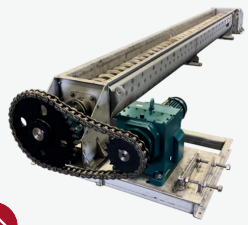
VERTICAL SCREW ELEVATOR

- Efficient vertical material elevation
- Requires uniform loading
- Designed with integral feeders
- Available with MTO features and stainless/special alloy construction



SHAFTLESS SCREW CONVEYORS

- Screw flight rides directly on liner, allowing for trough cleanout
- Ideal for handling materials with high moisture content
- No internal bearings



HEAT TRANSFER

- Transfers bulk material solids and manages heat transfer
- Available in "Core-Buster" and hollow-flight designs
- Constructed from various grades of steel depending on temperature needs
- Features large potential heat transfer surface area for high efficiency operation

PRECISION ENGINEERED
Built for consistent performance and reliability

BUILT TO LAST
Robust construction for demanding environments

CUSTOM SOLUTIONS
Tailored designs to meet your unique requirements

TRUSTED PARTNER
20+ years of delivering excellence worldwide

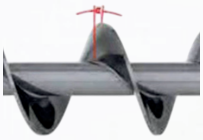
ENGINEERED FLIGHTS.
BUILT FOR PERFORMANCE.

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Precision engineered screw conveyor components designed for maximum efficiency, longer service life and reliable performance in the most demanding applications.



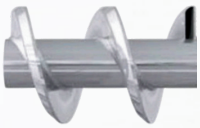
FLIGHT ACCESSORIES AND VARIATIONS



CANTED FLIGHT
Canted flights are used to counter the bending forces experienced when a flight is used in an extrusion process. The flight leans forward at a specific angle to the centre pipe.



COMPRESSING FLIGHT
Compressing flight centers reduce screw sweep volume, squeezing and compacting material as it moves forward. This continuous force makes the process ideal for compacting, dewatering, and decanting applications.



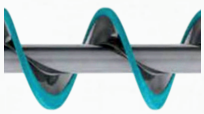
HOLLOW ELIX FLIGHT
Hollow flights are used to pass heating or cooling liquids around the helix. Typically both the hollow centre pipe and the hollow flight are used as a heat exchanger to cool, heat or thaw the material being conveyed.



PADDLES
Paddles are set in a helical form around a square or round centre tube to form a segmented spiral. They are used in mixers and pug mills to mix and condition dry products when liquid is added during the process.



WEAR SHOES
Bolt-on wear shoes are used as a sacrificial flight tip for conveying extremely abrasive products. Wear shoes significantly extend the working life of the spiral.



CERAMIC COATING
A matrix of ceramic coating is applied to the tip and leading face of the flight to reduce the wear caused by abrasive and corrosive materials being conveyed by the spiral.



BRISTLE EDGING
Bristle edging is a polymer brush spiraled onto the flight's trailing edge, cleaning residual product from the conveyor to prevent the breakage of delicate products like fertilizer prill.

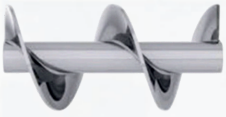


FLIGHT EDGING
Flight edging is an ultra high molecular weight polyethylene (UHMWPE) extrusion that is formed into a spiral and wraps around and snaps over the tip of the helix. Flight edging is used as an intermediate bearing on long slender spirals where hanger bearings are undesirable.



CENTRELESS DRIVE PLATE
A centreless drive plate is attached to a centreless spiral to enable the transmission of the driving torque to the spiral.

FLIGHT TYPES



- **STANDARD FLIGHT**
  Standard screws have a pitch spacing equal to their diameter and suit most conventional horizontal conveyors.



- **SHORT PITCH FLIGHT**
  Short pitch screws are used in inclined conveyors. They are also used to control the feed at the inlet and to reduce flushing of free-flowing materials. The shorter the pitch the more efficient is the screw.



- **VARIABLE PITCH FLIGHT**
  Variable pitch flights control material drawdown along a hopper. By increasing the pitch at each flight, you incrementally increase the volume of product entering the screw.



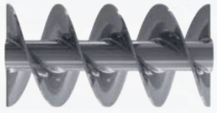
- **INTEGRABLE FLIGHT**
  Ribbon flights with integral support legs are used for sticky or cohesive materials that tend to build up on the flight-pipe interface.



- **NOTCHED FLIGHT**
  Notched flights are used to provide a gentle mixing action. Notched flights are useful in blending different materials during the conveying process.



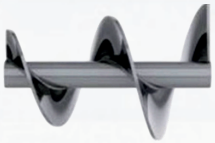
- **NOTCHED & FOLDED FLIGHT**
  Notched and folded flights provide aggressive mixing, effective aeration, and cooling by blending bypassed and trailing materials.



- **DOUBLE START FLIGHT**
  Double start flights provide conveyors with a gentle starting point, evening out the pulling centre for a smooth, uniform appearance.



- **CENTRELESS FLIGHT**
  Centreless flights are used for extremely sticky or fibrous materials that tend to wrap around or build up on a centre pipe.



- **TAPERED FLIGHT**
  Tapered flight spirals provide a continuous incremental change in swept volume, promoting even drawdown in full hoppers. However, they leave residual material in the conveyor unless the casing is tapered to suit.



- **CONED FLIGHT**
  Conical flights vary a screw's volume to ensure even hopper drawdown. Combining this coned screw with a variable pitch provides the optimal solution for uniform emptying. They are fully compatible with standard troughs.

PREMIUM QUALITY MATERIALS

PRECISION ENGINEERED

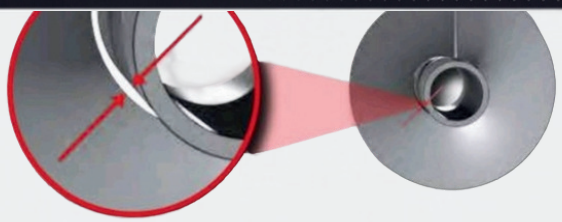
LONGER SERVICE LIFE & RELIABILITY

CUSTOM SOLUTIONS FOR EVERY NEED

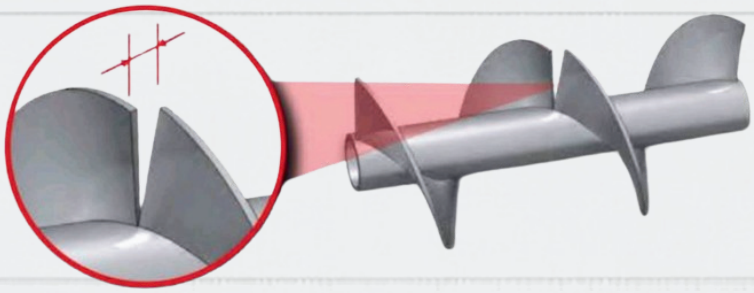
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COMMON FLIGHT PROBLEMS

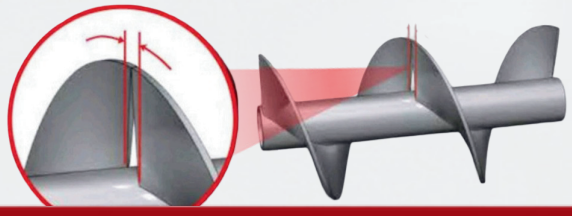
FLIGHTS ARE GAPPED AROUND PIPE



FLIGHT ENDS ARE MISALIGNED



FLIGHT ENDS ARE GAPPED



All these problems cost you time and money and result in an inferior product.

PROBLEM SOLVED

TRUHELIX STANDARD TOLERANCES

Outside Diameter 1%
Pitch ± 5%

Inside Diameter – It is expected that all flights click onto the centre pipe.

MATCH UP ALIGNMENT – It is expected that the misalignment where flights meet will be no more than half the material thickness.

MATCH UP GAP – It is expected that the gap where flights meet will be more than the material thickness.

PERFECT ALIGNMENT – Our screw flights are precisely aligned with the centre pipe at a perfect 90-degree angle all around, ensuring optimal balance, reduced wear, and improved efficiency in every application.

PRECISION ENGINEERED

IMPROVED PERFORMANCE

LONGER SERVICE LIFE

REDUCED WEAR & DOWNTIME



PRECISION. | ALIGNMENT. | PERFORMANCE. | DELIVERING EXCELLENCE.

AXIAL ENGINEERS
DELIVERING EXCELLENCE

PRECISION. PERFORMANCE. RELIABILITY.
Engineered Screw Conveyor Solutions for Every Need.

U-TROUGH SCREW CONVEYOR

Built to Convey.
Engineered to Perform.
Smooth Flow. Maximum Reliability.

Designed for efficient handling of powders, granules, flakes, and bulk materials, our U-Trough screw conveyors ensure smooth material flow, dependable operation, and long-lasting performance. Precision manufactured for industrial applications, they deliver reliable conveying solutions across diverse industries.



- Robust MS & SS Construction
- Continuous Material Transfer
- Low Maintenance Operation
- Custom Sizes & Capacities
- Heavy-Duty Industrial Design
- Tailor-Made Solutions Available

APPLICATIONS

- Food Processing
- Chemicals
- Fertilizers
- Cement
- Minerals
- Plastics
- Agriculture
- Pharmaceuticals

KEY FEATURES

- Efficient conveying of bulk materials
- Compact and space-saving design
- Consistent flow with minimal spillage
- Available in MS & Stainless Steel (SS)
- Custom lengths, diameters & capacities as per requirement

INCLINE SCREW CONVEYOR

Efficient | Reliable | Durable

Our Incline screw conveyors are designed for efficient transportation of powders, granules and bulk materials at an angle. Built for reliability, low maintenance and continuous operation across a wide range of industries.



KEY FEATURES

- Efficient conveying of bulk materials
- Compact and space-saving design
- Low power consumption
- Consistent flow with minimal spillage
- Available in MS & Stainless Steel (SS)
- Custom lengths, diameters & capacities as per requirement

APPLICATIONS

- Food Processing
- Chemicals
- Fertilizers
- Cement
- Minerals
- Plastics
- Agriculture
- Pharmaceuticals

HIGH EFFICIENCY
Smooth conveying with minimal material loss

LOW MAINTENANCE
Simple design for easy operation and minimum downtime

ROBUST CONSTRUCTION
Heavy-duty build for long life and reliable performance

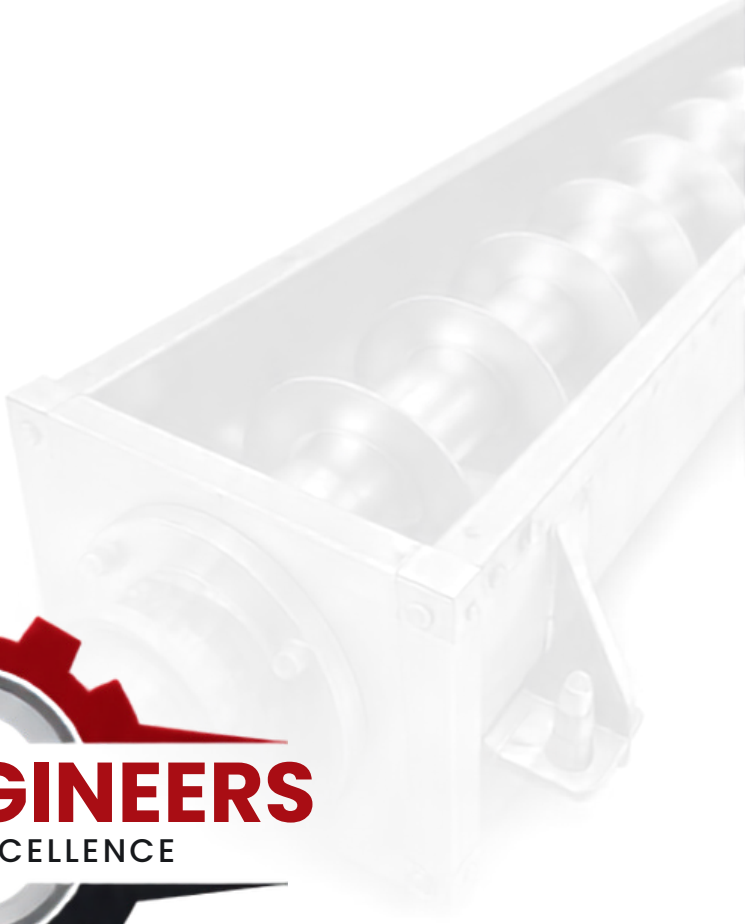
CUSTOM BUILT
Tailored sizes, capacities & materials to suit your process

BUILT TO LIFT.
BUILT TO PERFORM.

- RELIABLE PERFORMANCE
- LONG SERVICE LIFE
- BETTER EFFICIENCY
- MAXIMUM VALUE



20+ YEARS
OF MANUFACTURING EXCELLENCE
★ ★ ★



AXIAL ENGINEERS

DELIVERING EXCELLENCE

PRECISION ENGINEERED **BUILT TO LAST** DELIVERING EXCELLENCE

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